

Francisco Cardenas Beltran

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EDUCATION

University of California, Los Angeles

B.S. in Electrical Engineering — Computer Science Technical Breadth

Sep 2023 – exp. Jun 2027

Los Angeles, CA

Relevant coursework: Analog Circuits I & II, RF Electronics & Circuits, Introduction to RF and microwave Systems, Antenna Design I & II, Electromagnetics, Communication Systems, Digital Signal Processing, Control Systems.

PUBLICATION

J. Zhou, J. Du, C.-J. Tien, J.-W. Chen, R.-C. Soong, **F. Cardenas Beltran**, L. Cuskelly, C. Chen, M. Zhu, A. Tang, S.-W. Tam, and M.-C. F. Chang, "A 16-to-256QAM G-Band Subharmonic Phase-Modulating Transmitter for Beyond-5G Communications," IEEE International Solid-State Circuits Conference (ISSCC), 2026.

RESEARCH EXPERIENCE

High Speed Electronics Lab, UCLA (PI: Dr. M.-C. Frank Chang)

Los Angeles, CA

Student Researcher

Jun 2025 – Sep 2025

- Designed a series E-band integrated microwave power combiner in Keysight ADS (EM co-simulation), achieving <1 dB insertion loss across a 35 GHz passband using realistic circuit models, pole-frequency analysis, and Babinet's principle for the inductors.
- Supported on-chip testing and validation of a mmWave transmitter, coordinating probe measurements and data collection.
- Performed ground-signal-ground (GSG) probe landings and built automated test setups for repeatable mmWave measurements.
- Set up and debugged optical-table measurements with frequency synthesizers, oscilloscopes, waveguides, power meters, and power supplies.

Electron Loss Field Investigation (ELFIN), UCLA (PI: Dr. V. Angelopoulos)

Feb 2025 – Sep 2025

HF Loop Development Team Member Los Angeles, CA

- Optimized an electrically small loop antenna (radius, conductor thickness, copper weight) for sensitivity and cost in Keysight ADS, then verified performance with a network analyzer.
- Simulated toroids in ANSYS HFSS to optimize the balun transformer and selected toroids to maximize HF-loop sensitivity.
- Built MATLAB tools for SDR data acquisition and automated post-processing.
- Verified MIL-STD-461 radiated-emissions compliance for payload interfaces using low-frequency EM probes, spectrum analyzers, and MATLAB.

INDUSTRY EXPERIENCE

AnySignal — RF Communications Company

El Segundo/Torrance, CA

Hardware Engineering Intern

Jun 2026 – Sep 2026

- Designed 105-Watt SEPIC power converter
- Simulated layout of power stages both on test board as well as on product for stability and efficiency using ADS PE-Pro.
- Supported development of High Power Amplifier through test and measurement.
- Performed large signal characterization on PA's as candidates for 20 to 30 Watt High Power Amplifier product development.
- Developed scripts for testing for RF Power Amplifiers using PyVisa SCIPY commands as well as GNU Radio on a USRP B-200.

RF Technician / Electrical Engineering Intern

Jun 2024 – Sep 2024

- Troubleshoot transmission lines and power-supply circuitry to bring a multiband SDR within gain and efficiency specifications.
- Characterized products using spectrum analyzers, network analyzers, and power meters.
- Documented product repair and test procedures with the lead hardware engineer.

TECHNICAL PROJECTS

SEPIC Power Converter

July 2026 – Sep 2026

- Designed 105 Watt 22-36V to 32V, 3.2A power converter
- At 3A operation, achieved 85% efficiency as well as 1% voltage ripple
- Part selection followed EEE-INST-002

Integrated Microwave Power Combiner

Jun 2025 – Sep 2025

- Designed a series IC power combiner with sub-1 dB loss at D-band (125 GHz) to provide additional power-amplifier gain.
- Designed integrated transformer inductors using lumped circuit models and Babinet's principle, achieving 0.7 dB loss; met 28 nm CMOS layout constraints for a final combiner loss of 0.9 dB.

RF Transceiver Front-End (Wireless, RF, Analog Project) May 2025 – Jun 2025

- Designed a complete transistor-level RF transceiver front-end that successfully transmitted and received BPSK symbols.
- Simulated transceiver blocks in LTspice and performed impedance matching in Keysight ADS for peak power transfer.

LEADERSHIP

IEEE @ UCLA — Wireless Analog RF Project (WRAP) Lead Oct 2025 – Present

Project Lead Los Angeles, CA

- Lead hardware-design lectures on RF amplifiers, mixers, oscillators, and filters at the lumped-element and transistor level, plus RF fundamentals (P1dB, IP3, Smith charts, noise figure).
- Mentor 15 students through the design of analog/RF communication circuits in LTspice.
- Developed step-by-step lab guides for VNA and spectrum-analyzer measurements and documented reproducible procedures for characterizing transceiver blocks (gain, noise figure, linearity).

TECHNICAL SKILLS

Software & EDA: Keysight ADS, ANSYS HFSS, LTspice, QUCS, KiCad, MATLAB/Simulink, Python, C++

Hardware & Lab: Analog & RF IC design, RF transmission-line design, EMI/EMC, RF & noise-figure measurement, VNA, spectrum analyzer, frequency synthesizer